Individual Pathways

Activity 4.1

Revising chemical patterns

1. Identify the set that contains two metals and a non-metal.
   (a) Tungsten, niobium, bromine
   (b) Platinum, boron, neon
   (c) Fluorine, nitrogen, cobalt
   (d) Tin, sodium chloride, iodine

2. Identify the correct statement.
   (a) The number of neutrons in a nucleus always equals the number of protons.
   (b) When atoms lose electrons they become negatively charged.
   (c) Calcium atoms become stable when they lose two electrons to form a cation.
   (d) An anion has an excess of protons over electrons.

3. Select the correct statement.
   (a) There are two elements in period 2 of the periodic table.
   (b) The elements of group 2 of the periodic table have two outer shell electrons.
   (c) Elements of group 2 are more reactive than elements of group 1.
   (d) Lithium is a member of group 2.

4. Select the correct statement about uranium.
   (a) Uranium is a synthetic radioactive element.
   (b) Uranium has more electrons in its shells than Americium.
   (c) Uranium does not conduct electricity.
   (d) Uranium is a transition element that belongs to the actinoid series.

5. Select the statement that is true of arsenic.
   (a) Arsenic is a non-metal.
   (b) Arsenic is a member of group 4.
   (c) Arsenic is radioactive.
   (d) Arsenic is a metalloid.

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6. Chlorine and bromine are halogens. Which statement is true?
   (a) Bromine is a red-brown liquid and chlorine is a yellow-green gas.
   (b) Both are members of group 7.
   (c) Both are members of period 3.
   (d) Chlorine is a non-metal and bromine is a metalloid.

7. Which term best describes the ability of a metal to be bent or rolled into various shapes?
   (a) Lustrous
   (b) Ductile
   (c) Malleable
   (d) Brittle

8. What is the maximum number of electrons that can exist in the M shell?
   (a) 8
   (b) 18
   (c) 32
   (d) 50

9. A solution of a copper salt was sprayed into a Bunsen flame. What colour does the flame turn?
   (a) Green
   (b) Lilac
   (c) Yellow
   (d) Red

10. What is the chemical formula of aluminium oxide?
    (a) AlO
    (b) AlO_2
    (c) Al_3O_2
    (d) Al_2O_3

11. Bronze is an alloy of:
    (a) copper and zinc.
    (b) copper and nickel.
    (c) copper and tin.
    (d) tin and brass.

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12. The set that contains the metals in decreasing order of activity is:
    (a) Hg, Pb, Cr, Mn.
    (b) Hg, Cr, Mn, Pb.
    (c) Mn, Cr, Pb, Hg.
    (d) Cr, Mn, Pb, Hg.

13. The name of $KHCO_3$ is:
    (a) potassium carbonate.
    (b) potassium hydrogen carbide.
    (c) sodium hydrogen carbonate.
    (d) potassium hydrogen carbonate.

14. Answer true or false for each statement.
    (a) Calcium is an alkaline earth metal. ……..
    (b) There are 18 groups in the periodic table. ……..
    (c) There are 8 elements in period 1 of the periodic table. ……..
    (d) All members of group 1 have one valence electron. ……..
    (e) All members of group 18 have eight valence electrons. ……..

15. Answer true or false for each statement.
    (a) The modern periodic table is arranged according to atomic number. ……..
    (b) The electron configuration of neon is 2,8. ……..
    (c) The chemical formula of calcium sulfate is $CaSO_4$. ……..
    (d) Carbon monoxide is an ionic compound. ……..
    (e) Nickel sulfide is a covalent compound. ……..

16. Answer true or false for each statement.
    (a) Silver is more reactive than mercury. ……..
    (b) $CuFeS_2$ is a mineral called haematite. ……..
    (c) Sulfur trioxide is used to make sulfuric acid. ……..
    (d) In a nitrogen molecule, the two nitrogen atoms are linked by a triple covalent bond. ……..
    (e) The aqueous solutions of soluble salts conduct electric currents. ……..

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